Big O Order - Worked Solutions

Prepared Report

Question 1

Show that $f(n) = n^2 + 3n^3 \in O(n^3)$.

Step 1: Definition
We want constants $c > 0$ and $k > 0$ such that:
$0 \leq f(n) \leq c * n^3$ for all $n > k$.

Step 2: Factor the highest power of n
$f(n) = n^2 + 3n^3$
$= n^3 (3 + n^2/n^3)$
$= n^3 (3 + 1/n)$

Step 3: Bound the fraction
For $n \geq 1$, we know $1/n \leq 1$.
Thus: $3 + 1/n \leq 4$

Step 4: Final Inequality
$f(n) \leq 4n^3$ for all $n \geq 1$

Conclusion:
$f(n) \in O(n^3)$ with $c = 4$ and $k = 1$.

Question 2

What is the Big O order of $f(n) = n^2 + n \log_2 n + \log_2 n$.

Step 1: Identify growth rates
- n^2 dominates because it grows faster than $n \log n$ and $\log n$.

Step 2: Compare terms
For $n \geq 2$:
$n \log n \leq n^2$
$\log n \leq n^2$

Step 3: Simplify
$f(n) = n^2 + n \log n + \log n \leq 3n^2$

Conclusion:
$f(n) \in O(n^2)$ with $c = 3$ and $k = 2$.

Final Notes

Both solutions show the use of dominant term analysis and bounding to prove Big-O order.